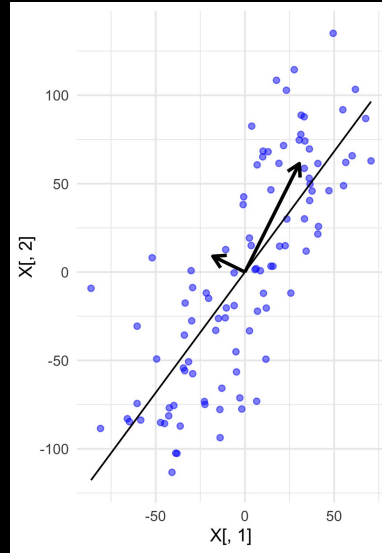


Linear Regression



Saket Choudhary

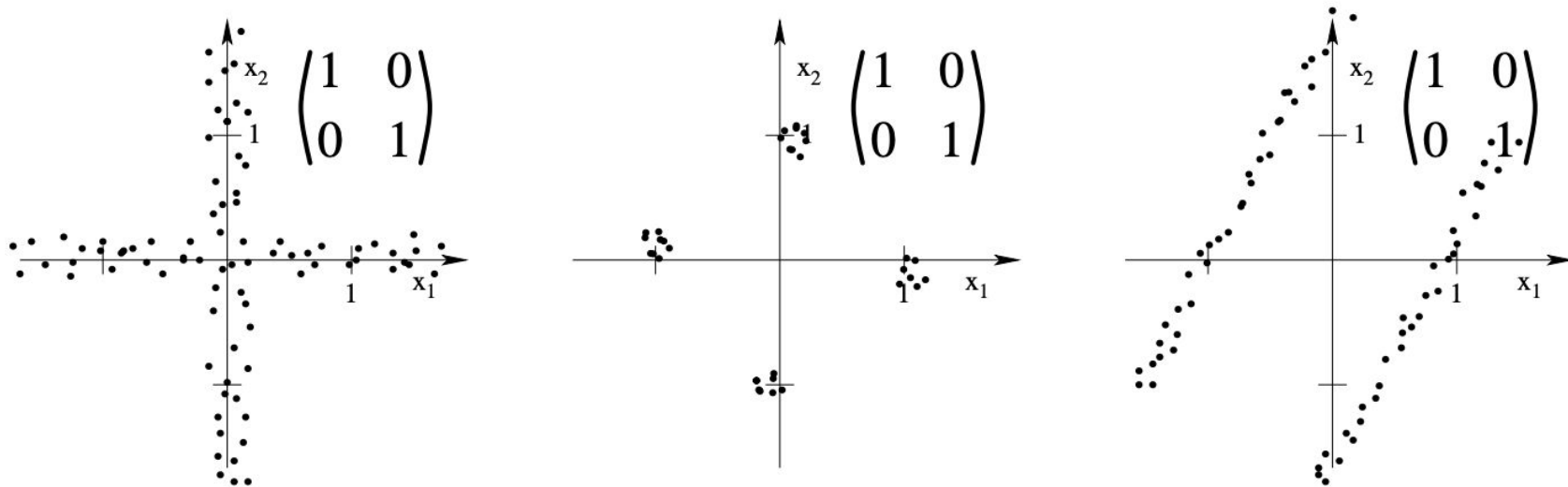
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Introduction to Public Health Informatics

DH 302

Lecture 10 || Friday, 7th February 2025

Covariance matrix captures the general extent of data



Different distributions with same covariance matrix

SVD Theorem

An arbitrary matrix $A \in \mathbb{R}^{m \times n}$ admits a decomposition of the form

$$A = \sum_{i=1}^n \sigma_i u_i v_i^T = U \tilde{S} V^T, \quad \tilde{S} := \begin{pmatrix} S & 0 \\ 0 & 0 \end{pmatrix},$$

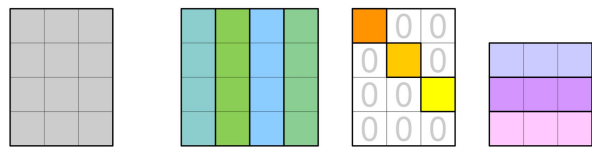
where $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ are both orthogonal matrices, and the matrix S is diagonal:

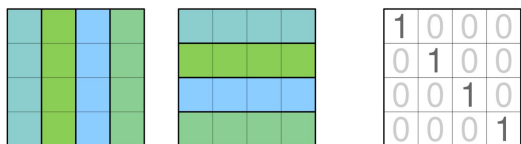
$$S = \mathbf{diag}(\sigma_1, \dots, \sigma_r),$$

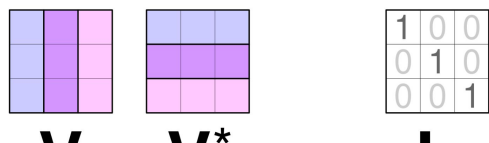
where the positive numbers $\sigma_1 \geq \dots \geq \sigma_r > 0$ are unique and are called the *singular values* of A . The number $r \leq \min(m, n)$ is equal to the rank of A , and the triplet $(U, \tilde{S}, V)$ is called a *singular value decomposition* (SVD) of A . The first r columns of U : $u_i, i = 1, \dots, r$ (resp. V : $v_i, i = 1, \dots, r$) are called left (resp. right) singular vectors of A , and satisfy

$$A v_i = \sigma_i u_i, \quad u_i^T A = \sigma_i v_i, \quad i = 1, \dots, r.$$

Singular value decomposition

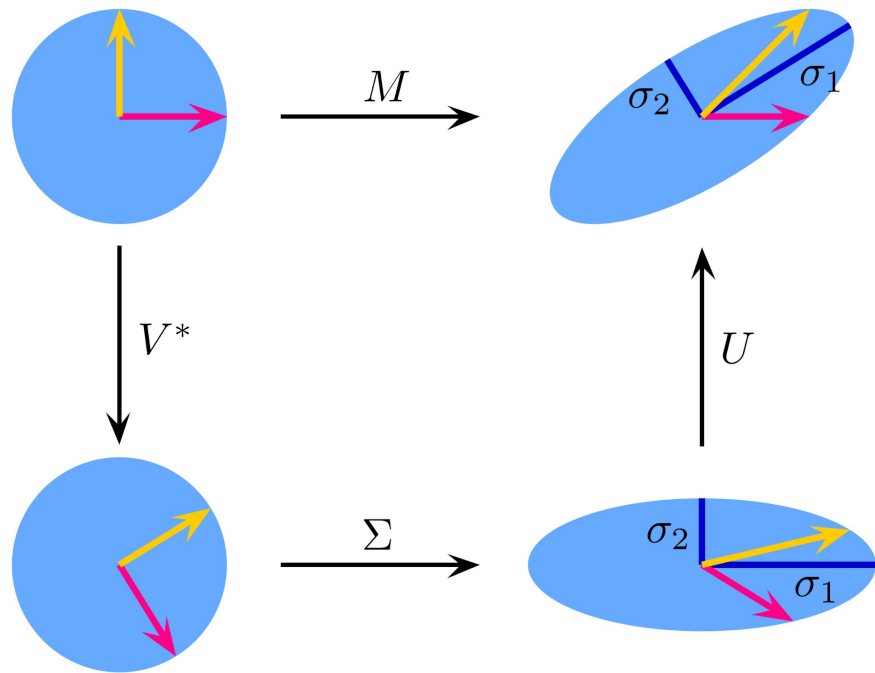

$$\mathbf{M}_{m \times n} = \mathbf{U}_{m \times m} \mathbf{\Sigma}_{m \times n} \mathbf{V}^*_{n \times n}$$


$$\mathbf{U} \mathbf{U}^* = \mathbf{I}_m$$


$$\mathbf{V} \mathbf{V}^* = \mathbf{I}_n$$

- Input: A matrix
- Output: a set of numbers called *singular values* and two collections of vectors: a set of *right singular vectors* and another set of *left singular vectors*.

Geometry of Singular value decomposition



$$M = U \cdot \Sigma \cdot V^*$$

- A given matrix M transforms the unit vectors into an ellipse
- This can be imagined as
 - 1. Performing rotation of the unit vectors by V
 - 2. Scaling these vectors by scaling factors (singular values of the matrix M)
 - 3. Performing another rotation by U

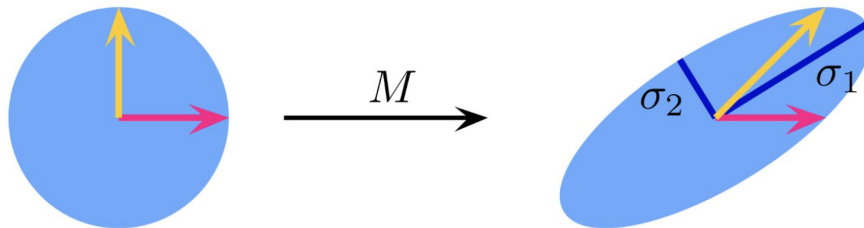
Singular value decomposition

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \quad \mathbf{\Sigma} = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & \sqrt{5} & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{V}^* = \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ -\sqrt{0.2} & 0 & 0 & 0 & -\sqrt{0.8} \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -\sqrt{0.8} & 0 & 0 & 0 & \sqrt{0.2} \end{bmatrix}$$

- The matrix M is rectangular.
- There are three non-zero singular values.
- The rank of M is 3
- $UU^T = I$ and $VV^T = I$ where I is the unit vector

What are singular values?

- A $m \times n$ matrix M can be thought of as mapping a vector x from R^n to R^m .
- A unit sphere in R^n is mapped to an ellipsoid in R^m :



- The non-zero *singular values* of M are the lengths of the *semi-axes* of the ellipsoid.

Eckart Young Theorem

- Let $M = U\Sigma V^T$ and set $M_r = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_r u_r v_r^T$
- Then M_r is the best low rank (r) approximation of M
- What does the theorem really say?
 - If you want a lower rank matrix, you can perform a SVD and retain the top singular values (and vectors)
 - Top singular values which measure the sizes of the largest of the semi-axes of the resulting ellipsoid from the transformation capture “most of the original information”

What is eigendecomposition of a matrix?

- A $n \times n$ square matrix A is diagonalizable if it can be written in the form $A = PDP^{-1}$ where D is a diagonal matrix, alternatively $AP = PD$

$$D = \begin{pmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{pmatrix} \quad P = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_n \\ | & | & \cdots & | \end{pmatrix}$$

- Decomposition of a square matrix A into the form $A\mathbf{p}_i = \lambda_i\mathbf{p}_i$ is called eigendecomposition as columns of P , i.e. $\mathbf{p}_i$ are eigen vectors of A and λ_i are eigenvalues

Singular values to eigenvalues (SVD to eigendecomposition)

- SVD of $M = U\Sigma V^T$. Consider $M^T M \rightarrow M^T M = (V\Sigma U^T)(U\Sigma V^T) = V\Sigma^2 V^T = V\Sigma^2 V^{-1}$
- $V\Sigma^2 V^T$ is the eigendecomposition of $M^T M$
- Thus, singular values of M = square root of the eigenvalues of $M^T M$
- Recipe (not best) to calculate SVD of a matrix M :
 - Compute $M^T M$
 - Compute eigenvalues of $M^T M$, σ = singular values of (M) = square root of eigenvalues of $M^T M$
 - Find eigenvectors of $M^T M \rightarrow V$
 - Find eigenvectors of $MM^T \rightarrow U$

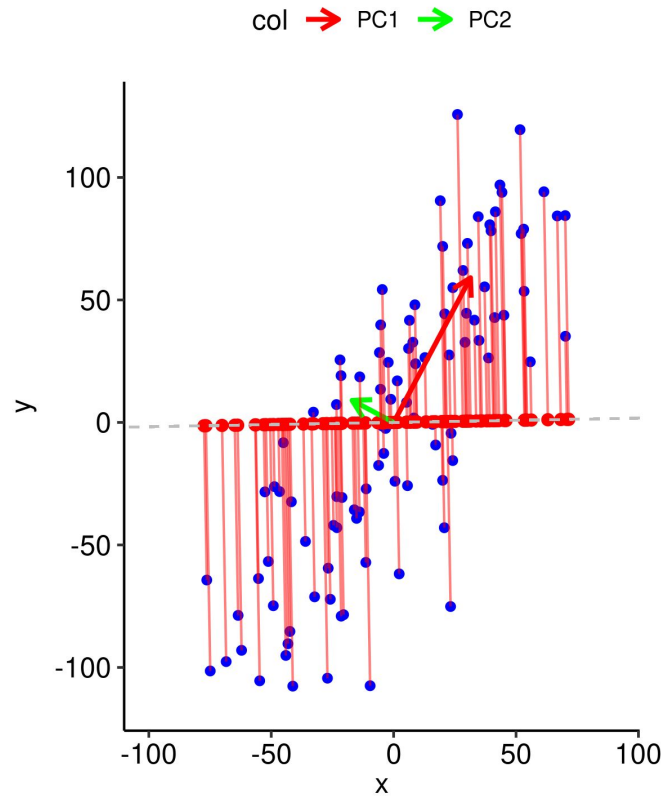
Eigendecomposition to SVD (eigenvalues to singular values)

- SVD is a generalization of eigendecomposition (which is only defined for square matrices)
- **Singular values** of a matrix M are the **square roots of the eigen values of MM^T**
- If M is a square matrix, with non-negative eigenvalues, then the above singular values and eigen values will be same

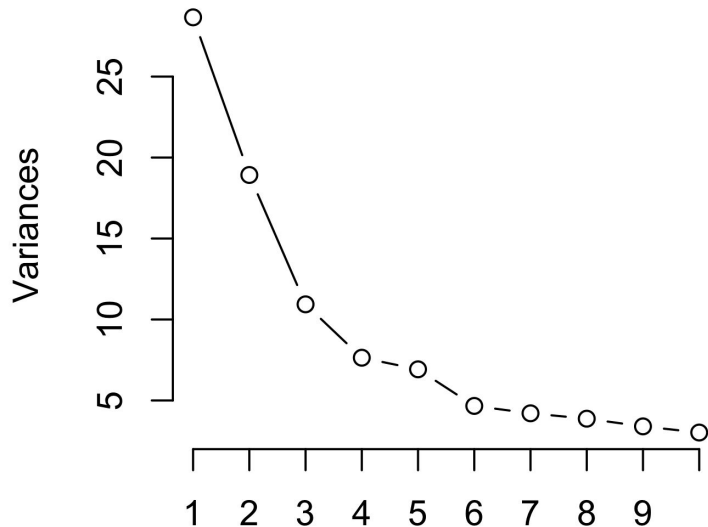
Principal Component Analysis - The recipe

- Start with a data matrix M .
- *Center* M by subtracting the column means (each column is a feature)
- Perform SVD of $M \rightarrow M = U\Sigma V^T$
 - U and V are orthonormal
 - Σ is a diagonal matrix of singular values
 - V is made of eigenvectors that diagonalize the covariance matrix $M^T M$.
- Truncate V_k to retain the first k columns
 - $M_k = U_k \Sigma_k V_k^T$ is a good low rank (k) approximation of M .
- “Project” the original matrix M onto V_k : MV_k
 - This projection has two properties:
 - It maximises the variance of projected points
 - It results in minimum reconstruction error if the original matrix is to be reconstructed

PCA: the optimization



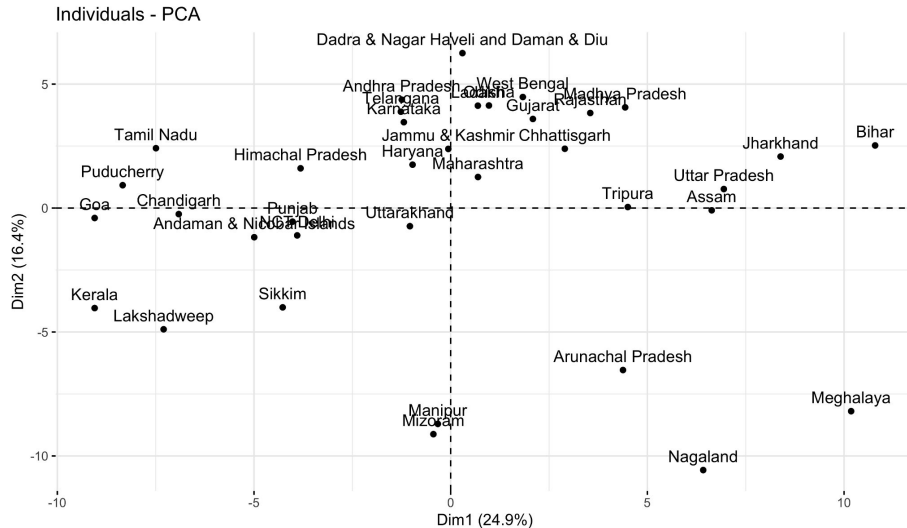
NFHS5 PCA reveals spatial clusters



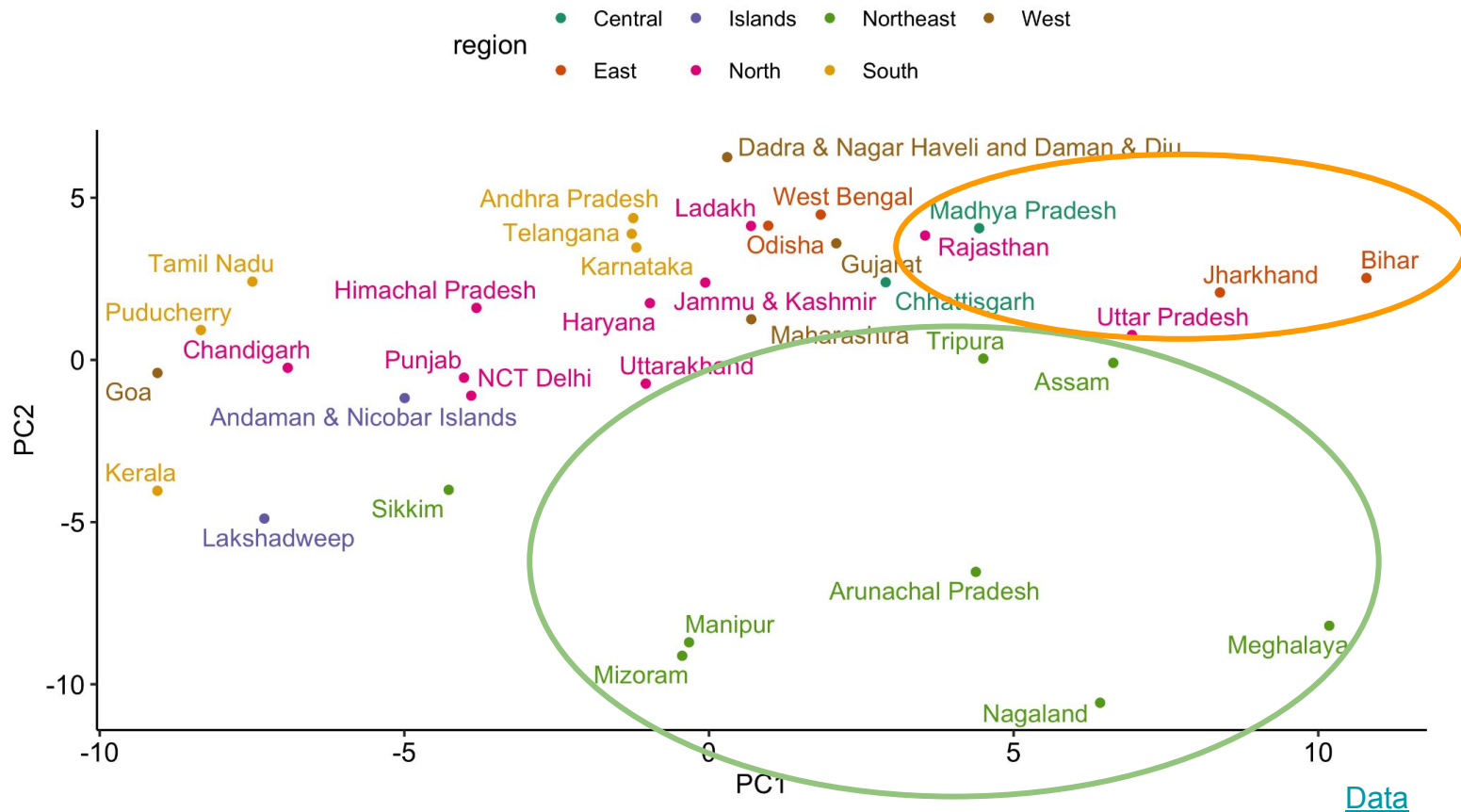
Variance explained by ith
PC = i-th eigen value

Percentage variance
explained = $100 \cdot \lambda_i / \sum \lambda_i$

```
pca <- prcomp(df, scale = TRUE)
df.pca <- as.data.frame(pca$x)
df.pca$state <- rownames(df.total.long.noindia.nona)
df.pca <- df.pca %>% select(state, PC1, PC2)
ggplot(df.pca, mapping = aes(x = PC1, y = PC2)) +
  geom_point() +
  geom_text_repel(aes(label = state), hjust = 0, vjust = 0)
# For plotting, alternatively
factoextra::fviz_pca_ind(pca)
```

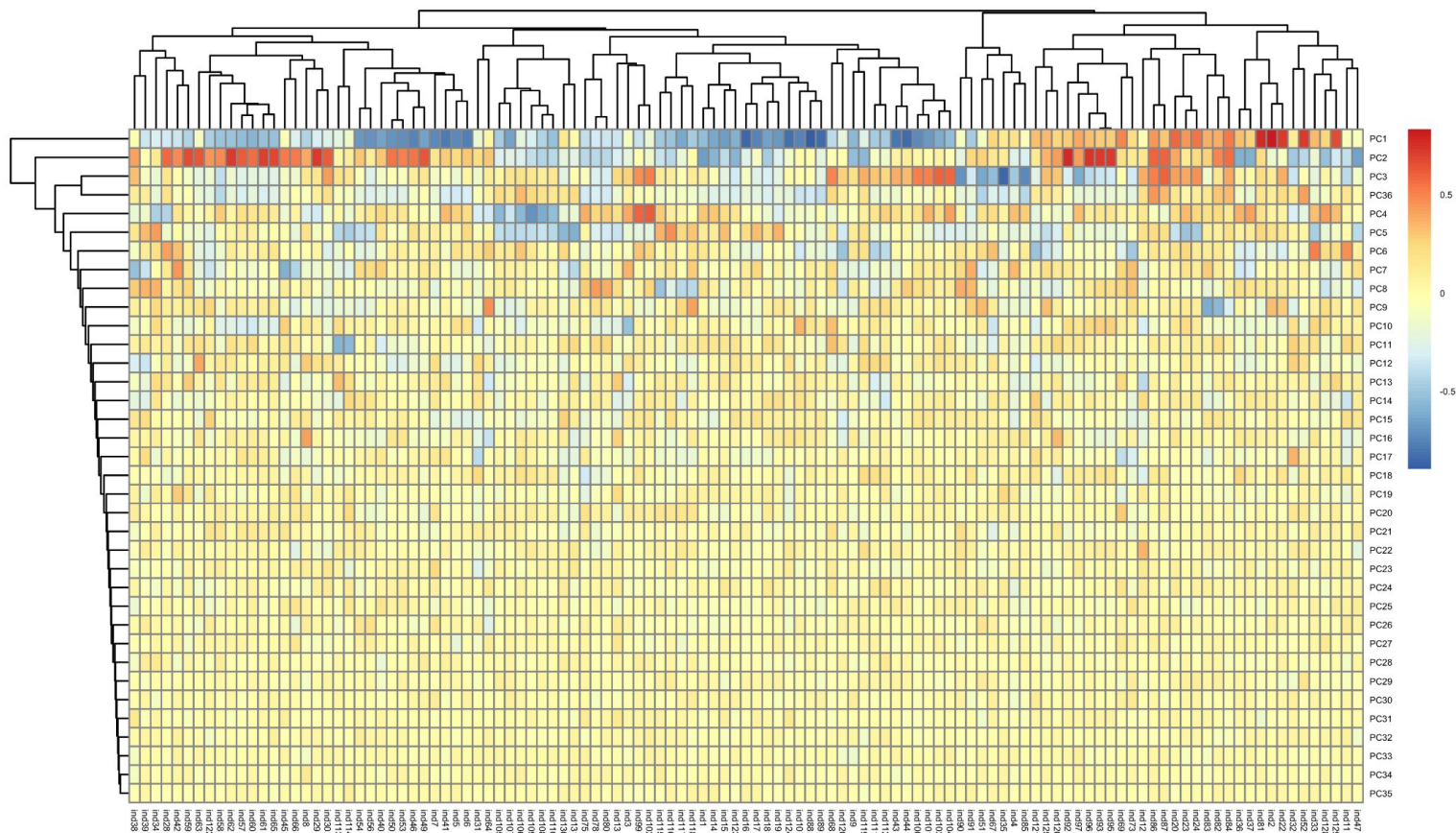


NFHS5 PCA reveals spatial clusters



NFHS5 PCA reveals spatial clusters

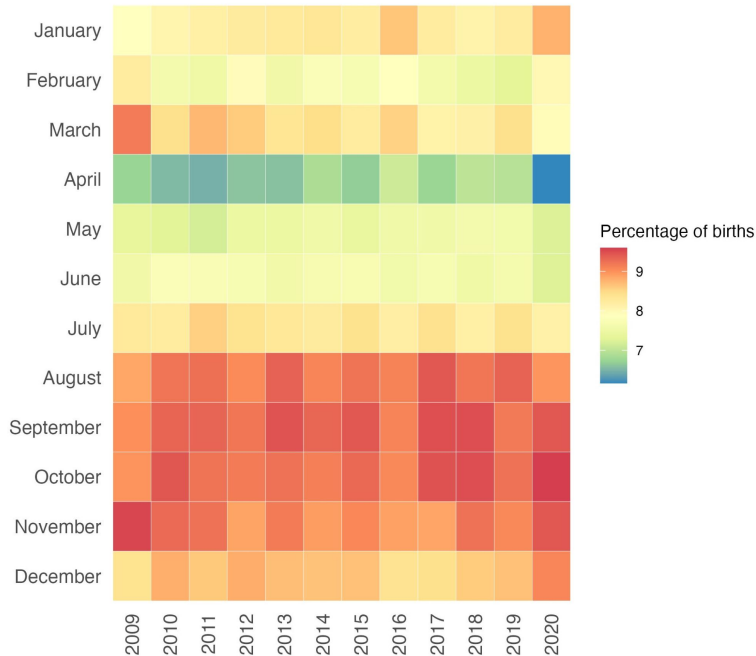
Correlation between PCs and original variables



Linear regression: Estimating relationships

Seasonality of births in India

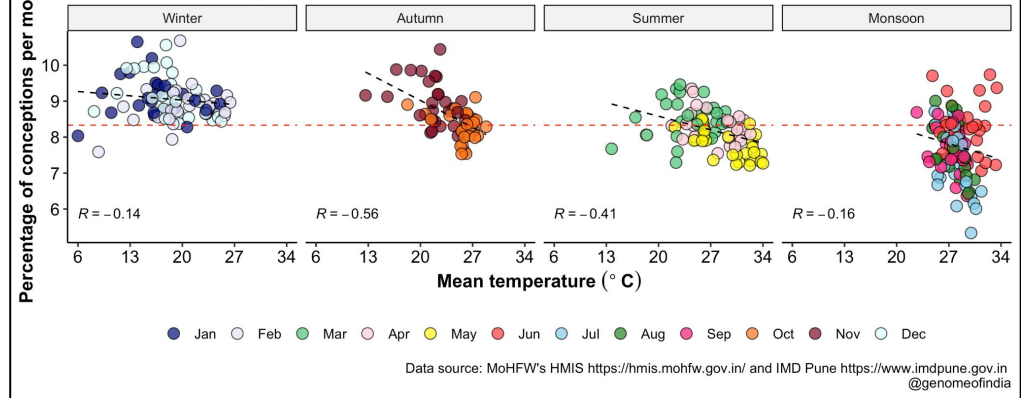
Percentage of live births per month in a year in India



Data source: MoHFW's HMIS database
@genomeofindia

Seasonality of conceptions in India

Correlation b/w percentage of estimated conceptions and temperature across seasons



<https://genomeofindia.substack.com/p/wake-me-up-when-august-ends>

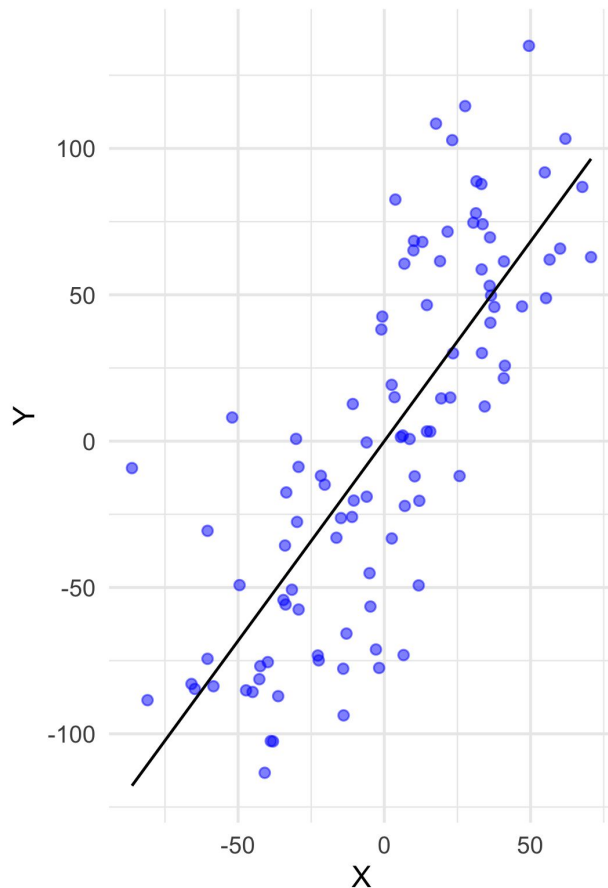
Linear regression

Regression analysis estimates the relationship between a dependent and one or more independent variable

Simplest form of regression analysis is **linear regression**: relationship between the parameters estimating the relationship between independent and dependent variable is linear

For a model to be linear, the parameters should be representable as a linear combination of the observables:

$$\hat{\beta} = \sum_i w_i y_i \text{ where the weights } w_i \text{ are estimated.}$$



Are the following models linear?

$$y = ax + b$$

$$y = ax^2 + b$$

$$y = ab + x$$

$$y = e^x \beta + c$$

$$y = e^\beta x + c$$

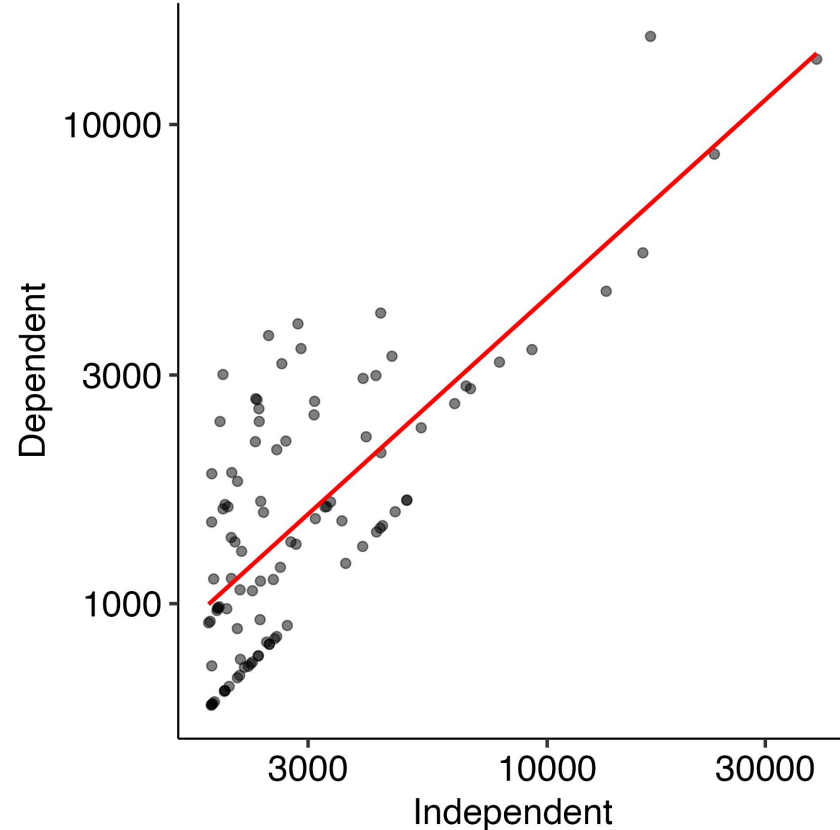
The “best fit” line

Given observations $\{\mathbf{x}_i, y_i\}_{i=1}^n$ in ordinary least squares (OLS) regression, we estimate the parameters β . Two possible approaches:

- Loss function based optimization $\rightarrow$ minimize the “prediction” error
- Distributional way: Assume the data is coming from a distribution and model it explicitly

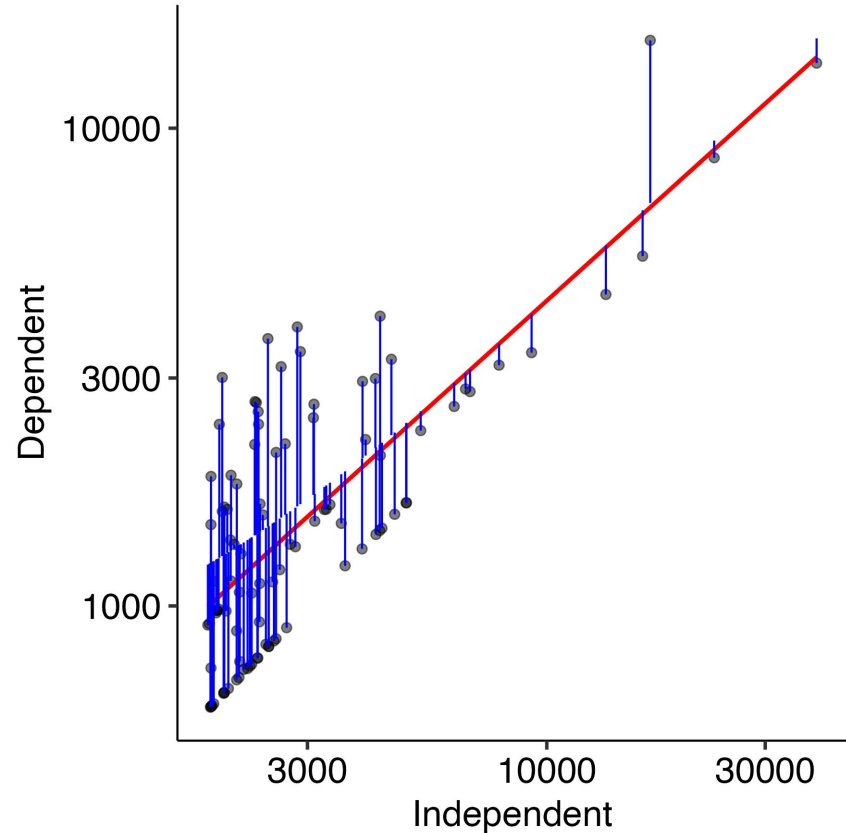
Linear regression: The Goal

- “Regression analysis” = Problem of estimating relationship between a **dependent variable** and one or more **independent variables**
- “Linear regression” = Simplest example of regression analysis where the relationship between the dependent and the independent variable is assumed to be linear in the form of parameters



Linear regression: How is the line fit?

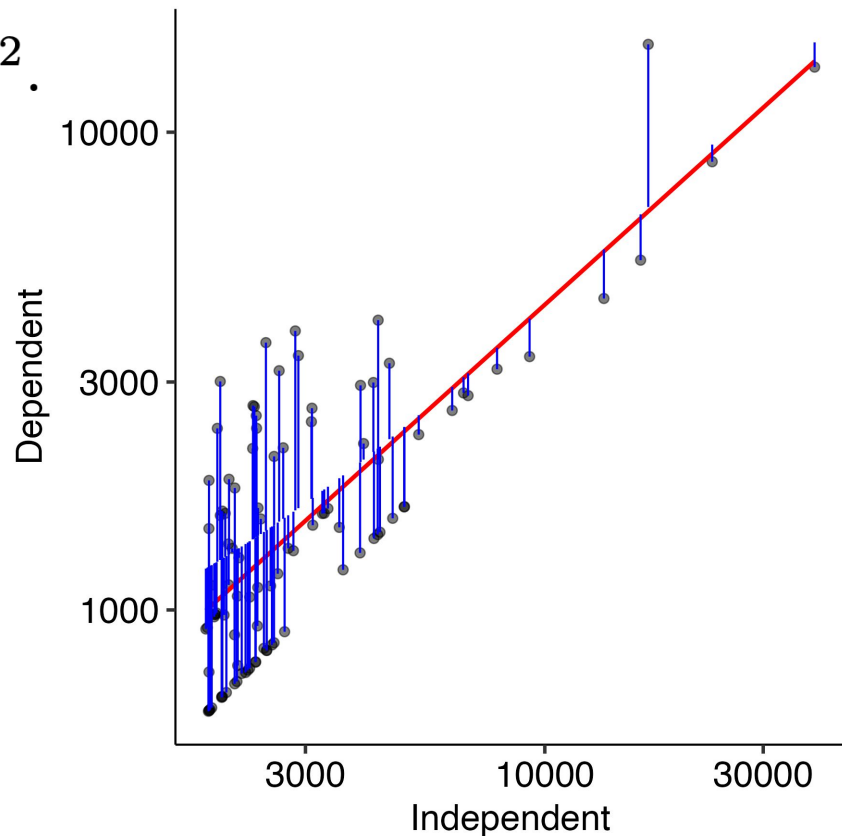
- **Goal:** Fit a line of the form $y = mx + b$.
 - M = Slope of the line
 - b = Intercept
- **Approach:** We find a line such that the sum of square distances (vertical) between the actual points and the line is minimized. Why **square**?
- Squared distances are “easier” to optimize



Fitting the line: Method 1 (multivariable calculus)

Minimize $f(m, b) = \sum_{i=1}^n (y_i - (mx_i + b))^2$.

- Take partial derivatives w.r.t m and b and set to zero
- Having a square here is helpful as the resulting derivative is easy (say compared to if it was an absolute value)



Fitting the line: Method 2 (linear algebra)

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \quad \mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,p} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{p,n} \end{bmatrix}$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \quad \hat{\mathbf{y}} = \begin{pmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{pmatrix} = \mathbf{X}\boldsymbol{\beta}$$

Minimize

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

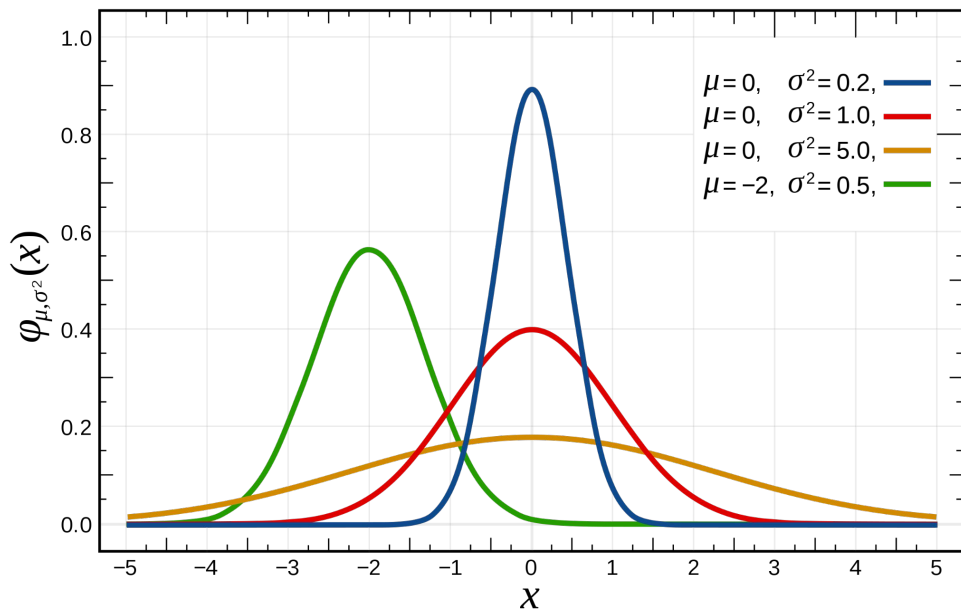
- $\mathbf{Y} = (y_1, \dots, y_n)^T$, is a point in an n -dimensional vector space
- Least squares optimization problem = Finding the nearest point on a subspace spanned by a column matrix $\mathbf{X}$ defined from the dependent variables.
- Invoking orthogonality, the solution emerges as $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$.

Fitting the line: Method 2 (Probabilistic)

- Normal (gaussian) distribution:

$$f(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}.$$

- Two parameters:
 - μ = mean
 - σ^2 = variance
- “Standard normal” has $\mu = 0, \sigma^2 = 1^2$.



Fitting the line: Method 2 (Probabilistic)

Independent variables (x) are measured without error, but there is error in the measurement of dependent variables (y)

$$y_i = mx_i + b + \epsilon_i$$

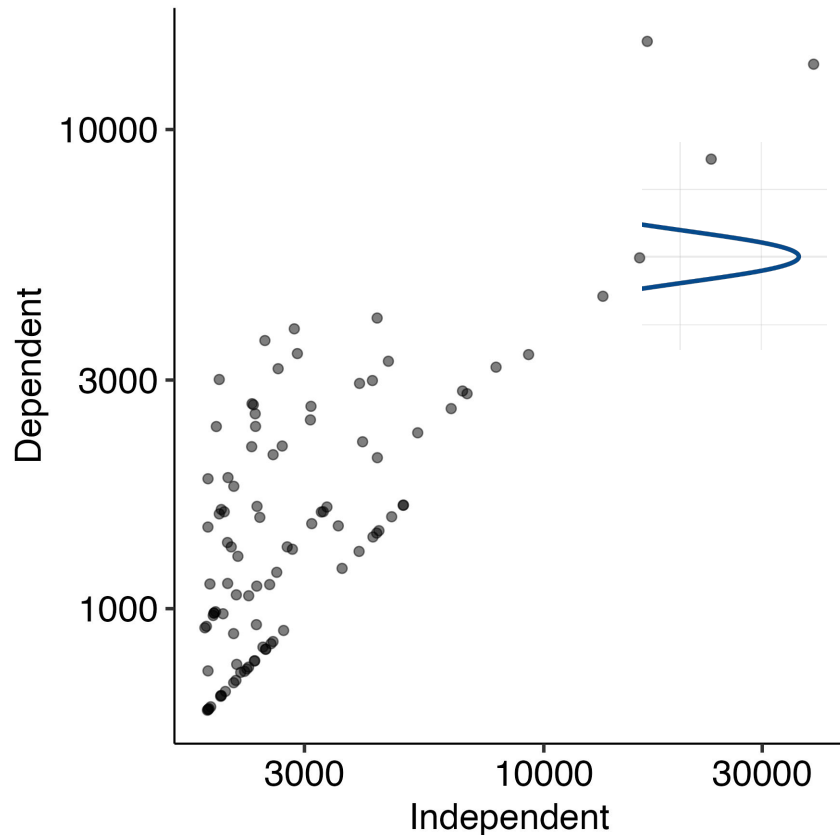
$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- Probability of observing y_i is then given by:

$$f(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (mx_i + b))^2}{2\sigma^2}}.$$

- The likelihood of observing data $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ is given by:

$$\mathcal{L}(m, b) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (mx_i + b))^2}{2\sigma^2}}.$$



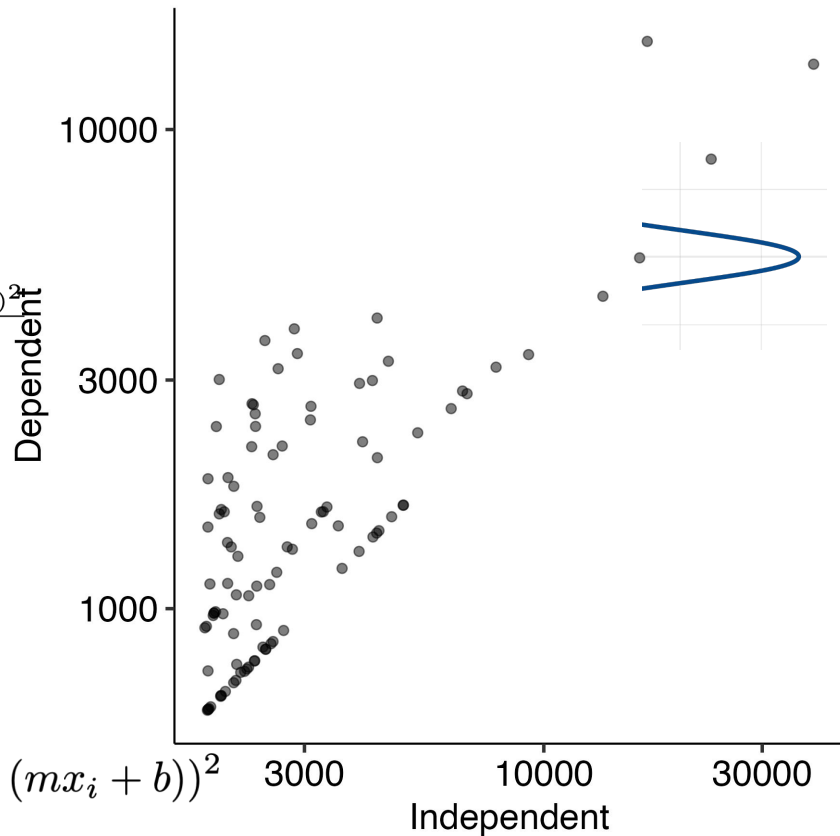
Fitting the line: Method 2 (Probabilistic)

- To find the “best fit” line, we would want to maximize the likelihood of observing the given data given the parameters

$$\operatorname{argmax}_{m,b} \mathcal{L}(m,b) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (mx_i + b))^2}{2\sigma^2}}$$

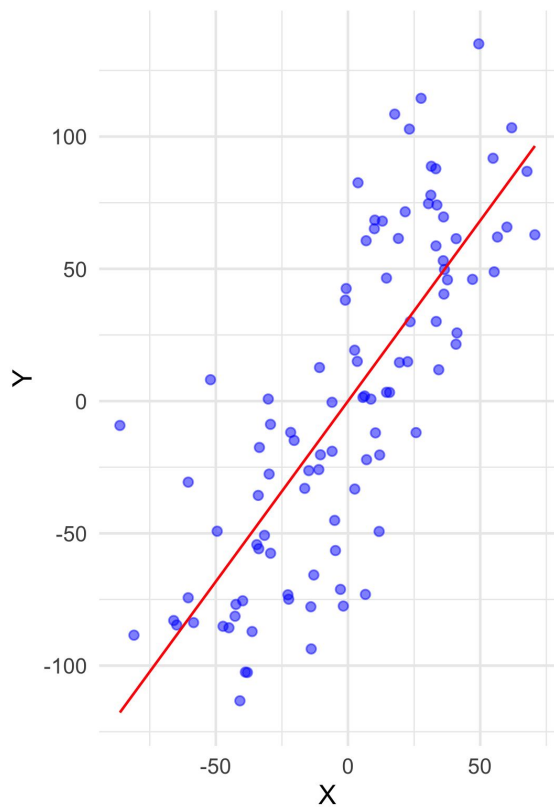
- Take the logarithm which is a monotonically increasing function and find the parameters m , b that maximize the logarithm of the likelihood function

$$\begin{aligned} \operatorname{argmax}_{m,b} \log(\mathcal{L}(m,b)) &= -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (mx_i + b))^2 \\ &= \operatorname{argmin}_{m,b} \sum_{i=1}^n (y_i - (mx_i + b))^2. \end{aligned}$$

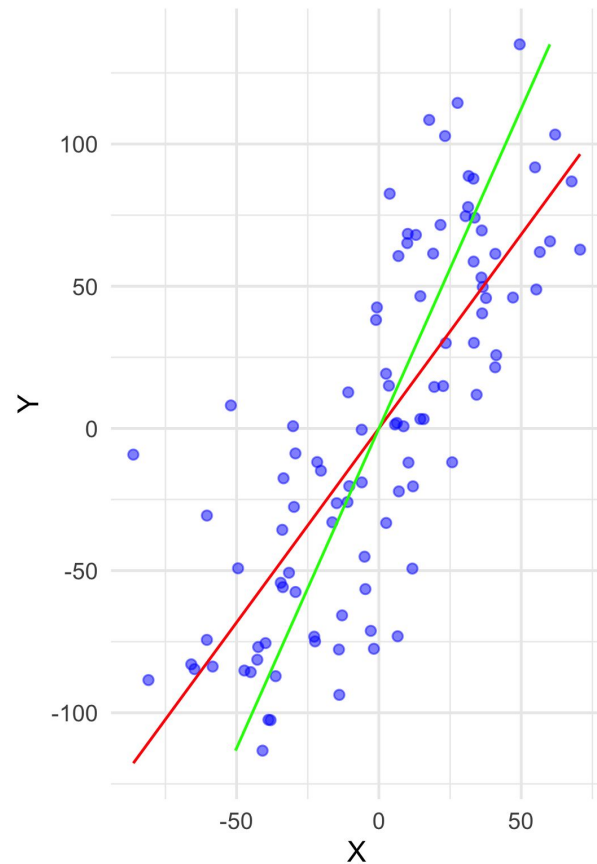
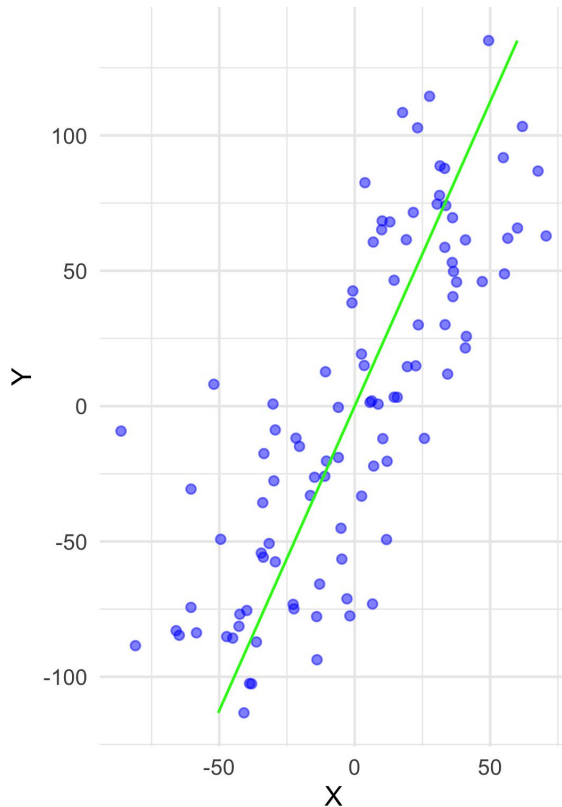


Dependent and independent variables should not be interchanged

`lm (y~x)`

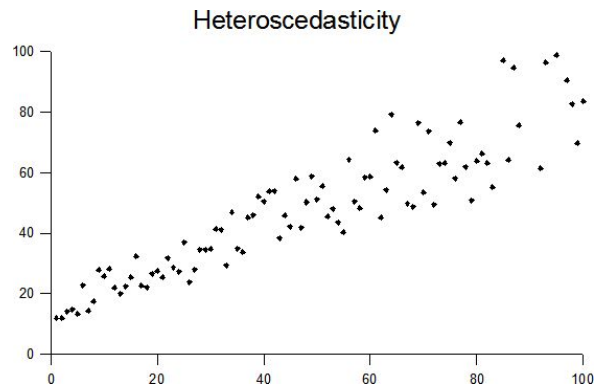
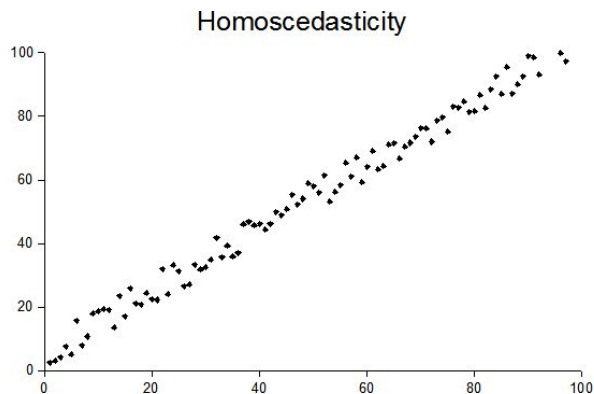


`lm (x~y)`



Homoskedasticity vs Heteroscedasticity

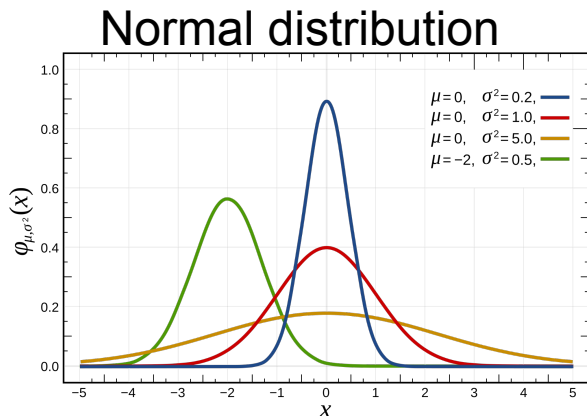
An inherent assumption in fitting a linear model is that of **homoscedasticity** → variance is constant across observations and is not a function of the observed quantity



What is so special about squares?

$$f(m, b) = \sum_{i=1}^n (y_i - (mx_i + b))^2$$

Least squared error fit can be interpreted as maximum likelihood for a linear model with gaussian noise in observations

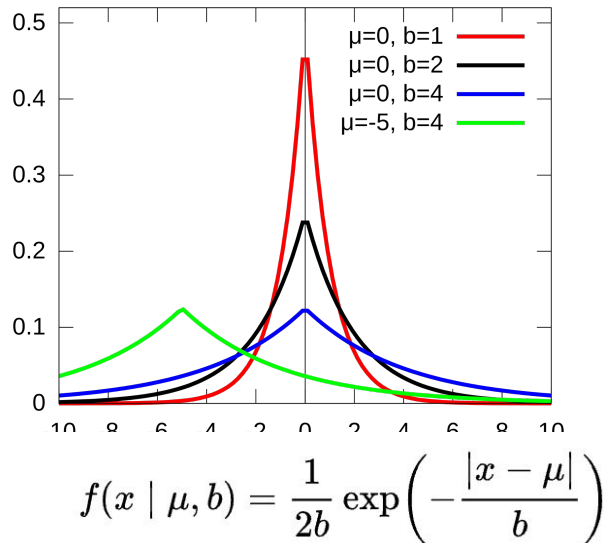


$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$f(m, b) = \sum_{i=1}^n |(y_i - (mx_i + b))|$$

Absolute fit can be interpreted as maximum likelihood for a linear model with laplacian noise in observations

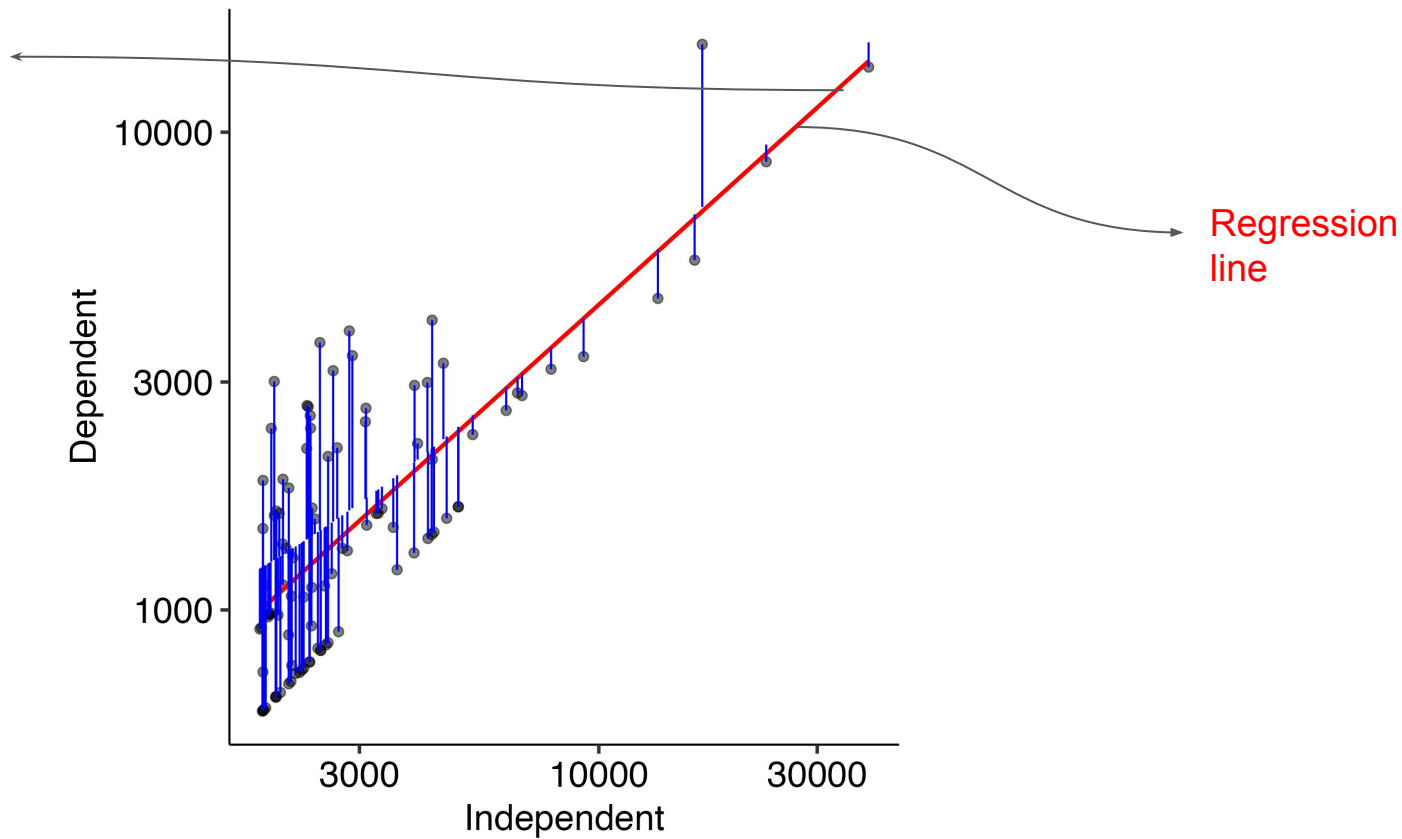
Laplace distribution



$$f(x | \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right)$$

The residuals

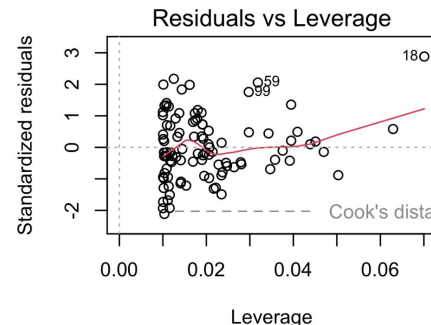
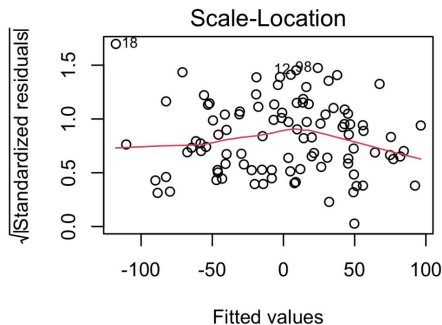
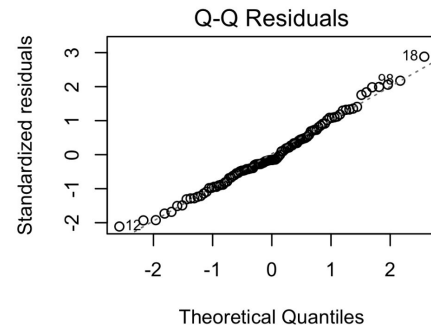
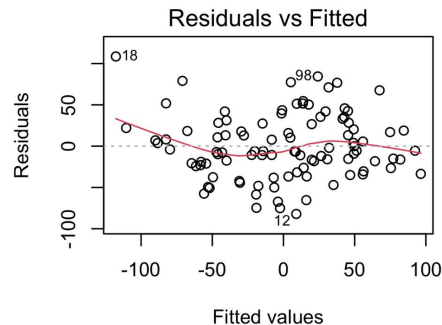
Residuals



Diagnosing the fit

- *Residuals vs Fitted* - Do residuals have non-linear patterns?
 - Good: Equal spread across fitted values
 - Bad: Some distinctive pattern (increasing/decreasing/U-shaped)
- *Q-Q* - Are the residuals normally distributed?
 - Good: Straight line (normality assumption is satisfied)
 - Bad: Strong deviations from normality
- *Scale Location* - are residuals spread evenly along ranges of predictors (check for homoscedasticity)
 - Good: Equal spread across fitted values
 - Bad: Some distinctive pattern (increasing/decreasing/U-shaped)
- *Residual vs Leverage* - Identify “influential” cases:
 - Good: Most points in $[-2,2]$ of y axis (standardized residuals)

```
linear_fit <- lm(y~x, data = as.data.frame(X))  
par(mfrow = c(2, 2))  
plot(linear_fit)
```



Running linear regression in R and diagnostics

Cook's Distance: Used to measure influence of a data point. How different would predictions be if the i-th point were to be deleted?

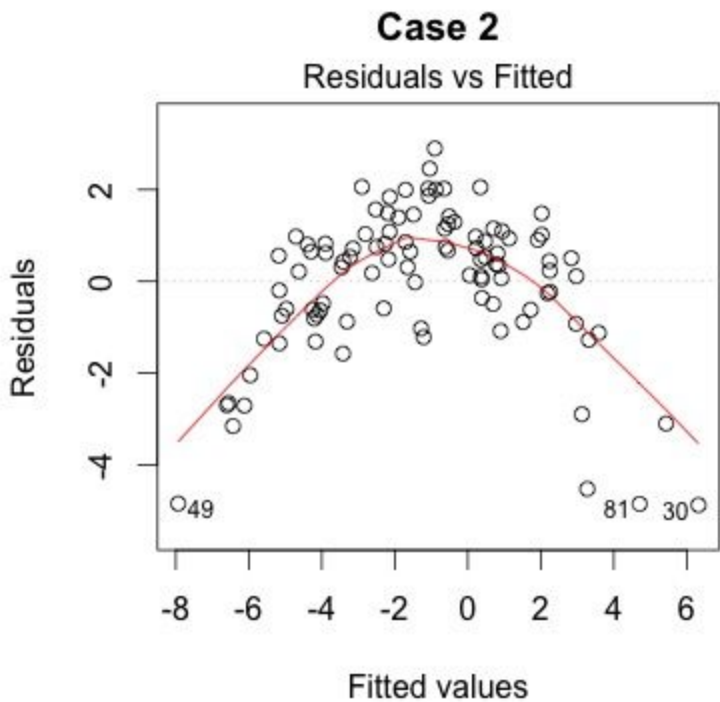
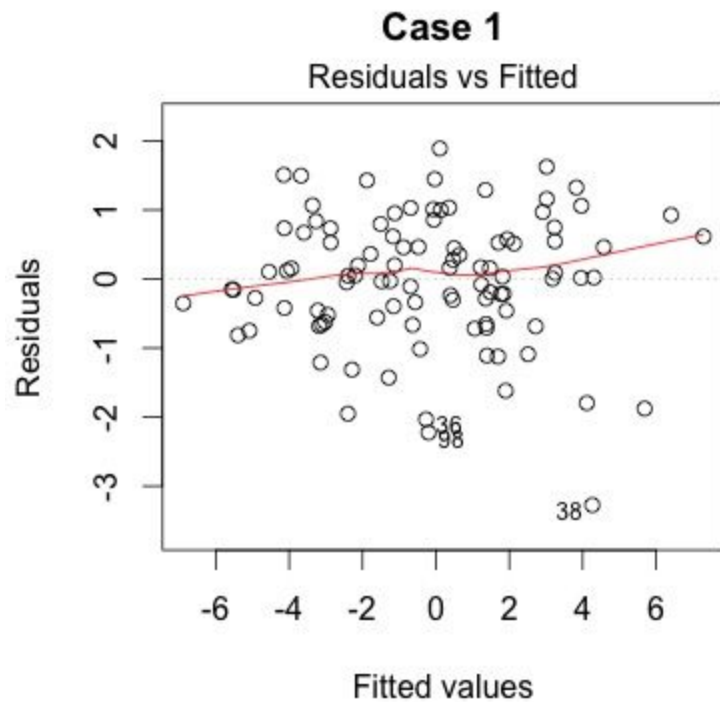
$$D_i = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(i)})^2}{ps^2}$$

P = rank of model

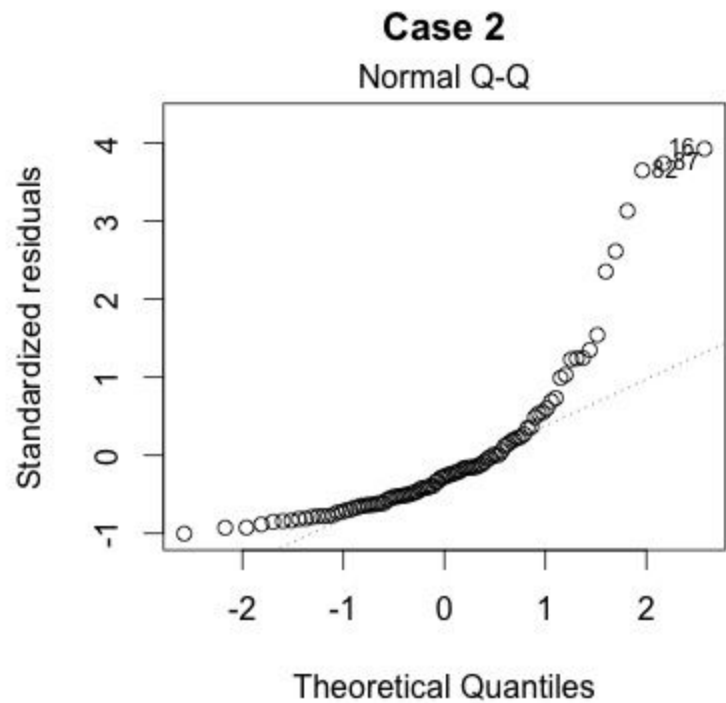
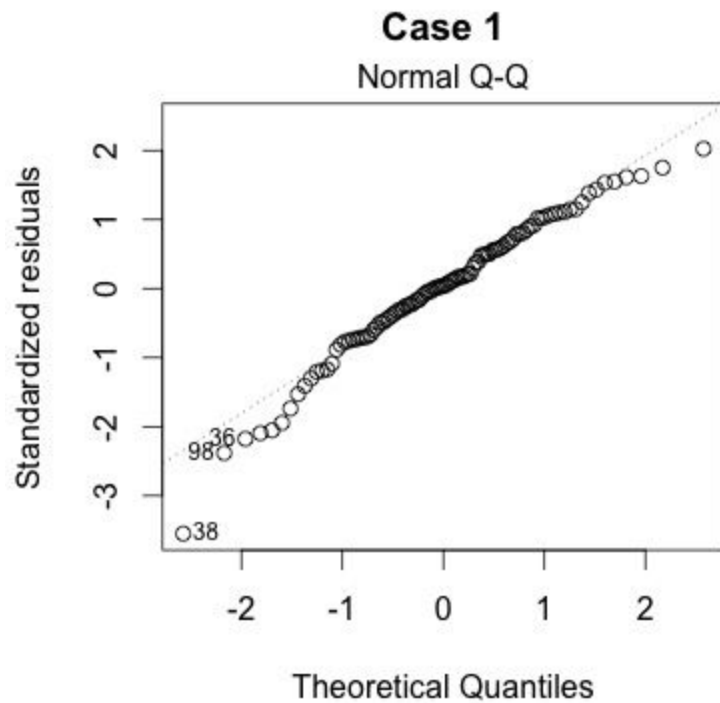
s² = Squared error of the model

$\hat{y}_{j(i)}$ Fitted y value when i-th observation is deleted

The good vs the bad



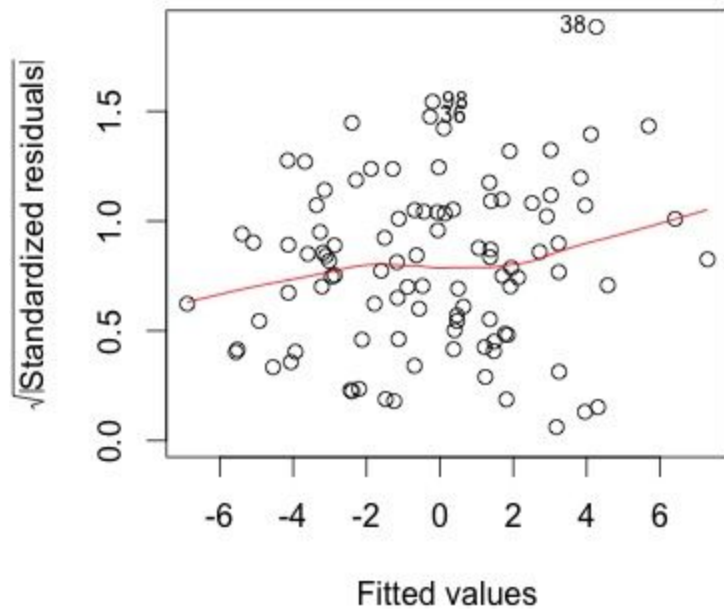
The good vs the bad



The good vs the bad

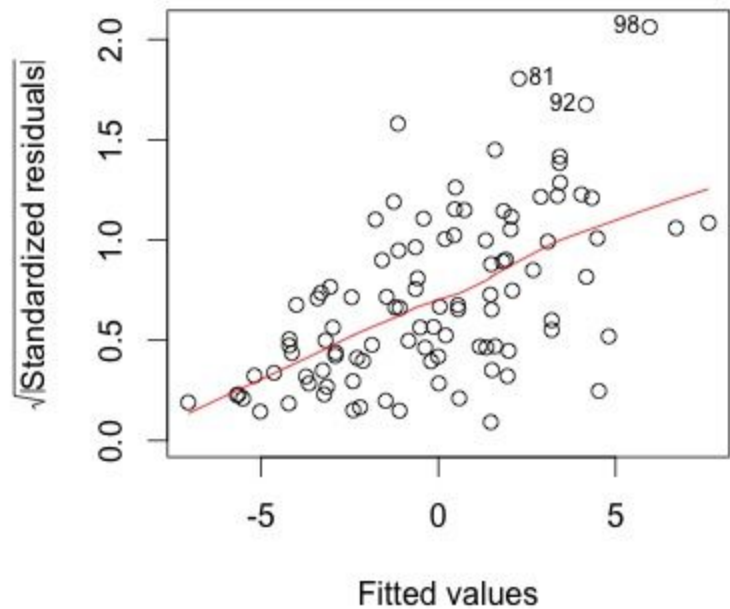
Case 1

Scale-Location



Case 2

Scale-Location



The good vs the bad

